Differential Equations

Solving by separation of variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given that $y = 3$ when $x = 0$.

2 Solve $\frac{dy}{dx} = 2xy$, given that $y = 5$ when $x = 0$.

Modeling with differential equations

3 A population P grows at a rate proportional to its current size, modeled by $\frac{dP}{dt} = kP$. If the population doubles every 5 years, find k, to three decimal places.

4 Newton's law of cooling states $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve this differential equation for $T(t)$ in terms of the initial temperature T_0.

First-order linear differential equations

5 Solve $\frac{dy}{dx} + 2y = 6$ using an integrating factor, given $y = 1$ when $x = 0$.

Euler's method for numerical solutions

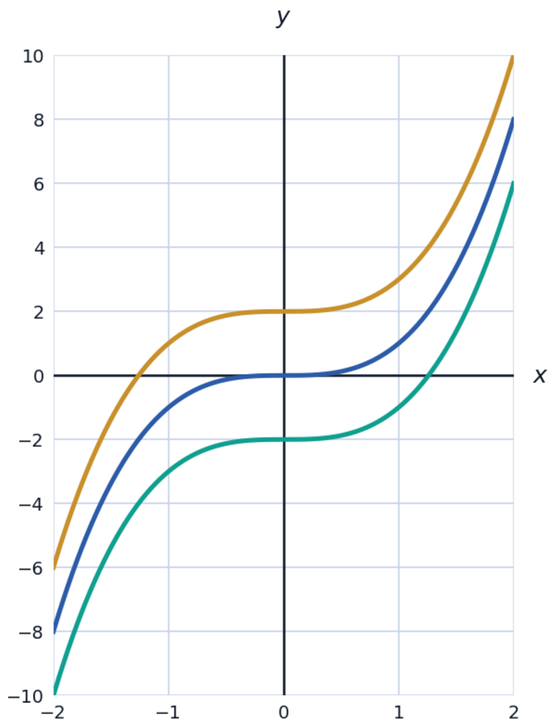
6 Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for $\frac{dy}{dx} = x + y$, given $y(0) = 1$, showing each step.

Second-order context: simple harmonic motion

7 The differential equation $\frac{d^2x}{dt^2} = -\omega^2 x$ describes simple harmonic motion. Verify that $x(t) = A\cos(\omega t)$ is a solution.

Family of solution curves and particular solutions

8 Find the general solution of $\dfrac{dy}{dx} = 3x^2$, and explain what the arbitrary constant represents geometrically.



9 A tank drains such that $\dfrac{dV}{dt} = -k\sqrt{V}$ for some constant $k > 0$, where V is the volume of water. Solve this differential equation for $V(t)$, given $V(0) = V_0$.

Verifying a solution to a second-order differential equation

10 Verify that $y = e^{2x}$ is a solution of the differential equation $\dfrac{d^2y}{dx^2} - 5\dfrac{dy}{dx} + 6y = 0$.

Slope fields and qualitative solutions

11 A differential equation is given by $\dfrac{dy}{dx} = x - y$. Find the gradient of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, and explain what a slope field represents.

Solving a linear differential equation with an initial condition

12 Solve $\dfrac{dy}{dx} - \dfrac{y}{x} = x$ for $x > 0$ using an integrating factor, given $y = 2$ when $x = 1$.

A logistic growth model

13 A population P grows according to the logistic differential equation $\frac{dP}{dt} = kP(10 - P)$. Using the partial fraction decomposition $\frac{1}{P(10-P)} = \frac{1}{10P} + \frac{1}{10(10-P)}$, solve for P in terms of t, given $P(0) = 1$ and leaving k as an unknown constant.

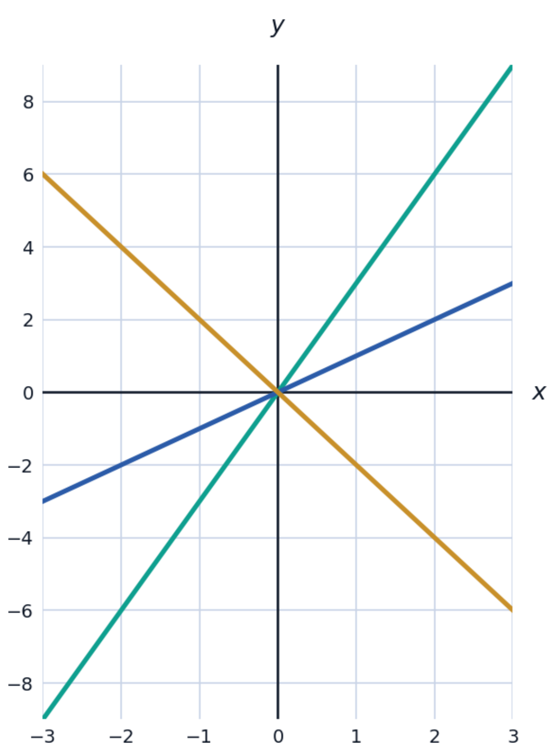
Checking the order and degree of a differential equation

14 State the order of each of the following differential equations: (a) $\frac{dy}{dx} = 3x^2 - 2y$, (b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$, (c) $\frac{d^3y}{dx^3} = \sin x$.

Using Euler's method with a smaller step size

15 Use Euler's method with step size $h = 0.25$ to estimate $y(0.5)$ for $\frac{dy}{dx} = 2x$, given $y(0) = 1$, showing each step, and compare with the exact solution $y = x^2 + 1$.

16 Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$, and describe the family of curves this represents.



17 A cup of coffee at 90° C is placed in a room at ambient temperature 20° C. Using Newton's law of cooling, $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 10 minutes the coffee has cooled to 60° C, find the value of k to three decimal places.

18 This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let $S(t)$ be the amount of salt (in grams) in the tank at time t minutes. (a) Explain why $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 0$. (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.